==

PlayList : spring boot full course

channel : india free internship

youtube channel [*https://www.youtube.com/@IndiaFreeInternship*](https://www.youtube.com/@IndiaFreeInternship)

source Code: *spring boot source Code*

==

SESSION 1

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1. SPRING FRAMEWORK
2. SPRING BOOT
3. SRPING DATA JPA
4. SPRING WEB MVC
5. SPRING REST
6. SPRING SECURITY
7. MICROSERVICE
8.SPRING BATCH
9.SPRING SCHEDULAR

=====================================

SESSION 2

=========

Technolgies : webPage(servlets) +Database(JDBC/JPA) + Email (Java Mail) + Message(JMS)

Limitations:

- *Start coding from line zero*
- *Development takes more time*
- *More chance of error*
- *Design pattern applied manually*

Framework:

- *Ready-made structure*
- *In-built technologies*
- *Design patterns already implemented*

RAD => Rapid Application Development

- *Fast coding*
- *Less Error*

Spring Framework:

- *used for Web Application or Distributed appliation*
- *It has own Apis*
- *It Follow Container System*

Spring Container

- *-Find/Scan Classes[spring bean]*
- *Create Object*
- *provide DATA*
- *-Link Object based on Relation*
- *Finally Destroy*

Depedency Injection (DI)

Depedency Injection is a concept/theory which is implemented by spring IoC (Inversion of control) also called as spring container.

Theory	programming
oops	core java
ORM	HIbernate with JPA
DI	Spring Ioc(Inversion of control)

---

session 3
=========

Application
=====================

3 Type of application

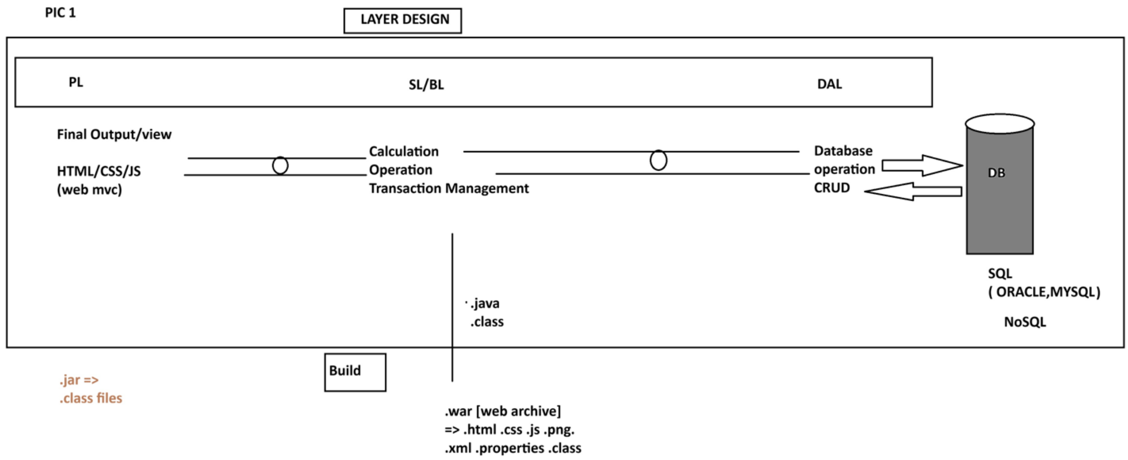
(i) Web Application
(ii) Distributed Application
(iii) Microservices Application

(i) Web Application : Collection of web pages run under webserver.

Application : - n services
Project :- n Module

eg : Gmail Application
User(Register and Login) , Inbox, Sent, Drfats, Setting, ... etc
Amazon Application
User(Register and Login) , search, Cart, Payment, ...etc
=> Implementation of service / module: - This is implemented using 3 Layers Design.
1. DAL : Data Access Layer (Database Operations)
2. SL : Service Layer (Calculation/Operations/txn mngmt)
3. PL : Presentation layer (view /Dynamic web pages/MVC)

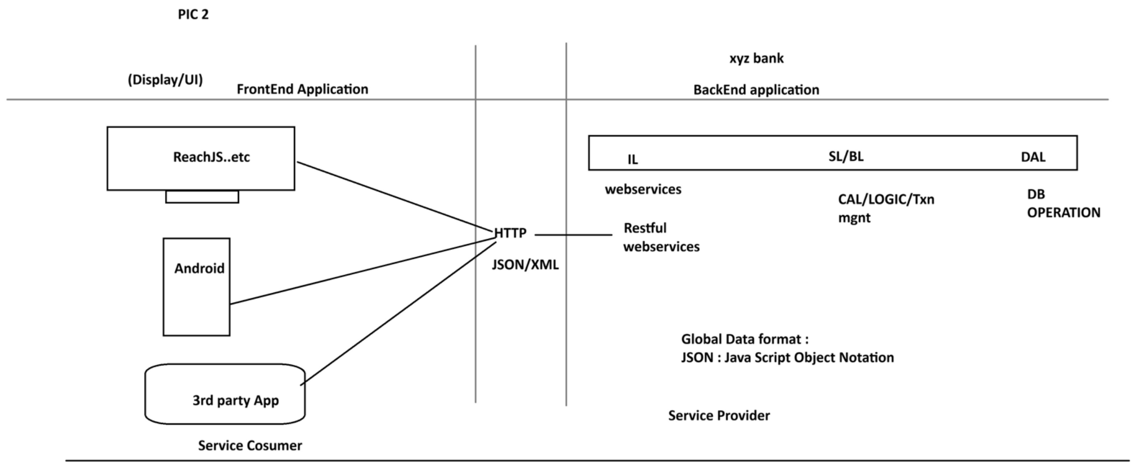
PIC 1



===

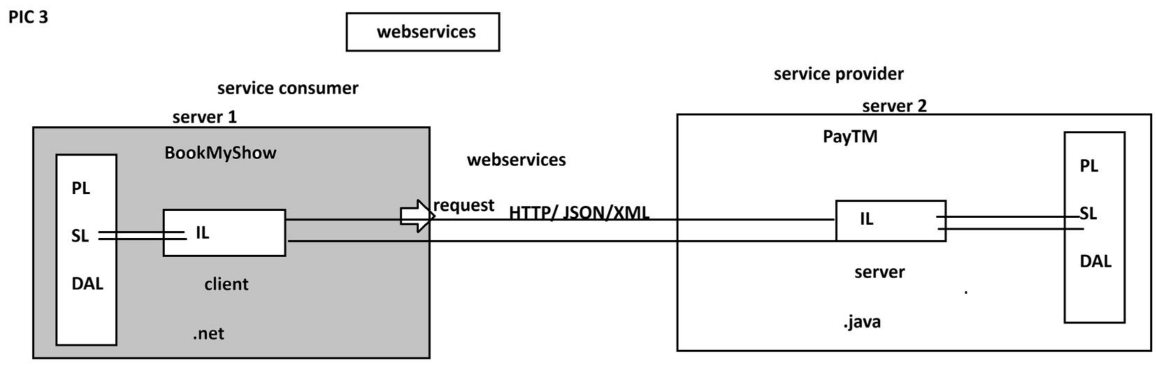
(ii) Distributed Application : An Application that runs at mutiple devices is called as distributed appliation.
xyz Bank application
This appliation can be accessed in different ways. Example.
eg:
1. Manually Process
2. NetBanking (Web App)
3. Mobile Banking
4. IVR
5. ATM / DC
6. 3rd party app (Gpay, PayPhone..etc)
.....

-But here business logics exist at backend application.
- Fontend applications are implemented using Angular, Android...etc
- App connected using HTTP Protocol using Global Data Format (JSON/XML)
PIC 2



WebServices:
===========
Intergration of multiple (at least two) applications which runs on different servers and implemented using different language (java, .net, php, python....etc)

PIC 3


PIC 3
webservices
service consumer
server 1
BookMyShow
PL
SL
DAL
IL
client
.net
webservices
request
HTTP/ JSON/XML
service provider
server 2
PayTM
IL
server
.java
PL
SL
DAL

==

session 4

================
================
SPRING FRAMEWORK
================
CORE
----
1. DI and IOC
2. Container Types
3. Annotation Configuration
4. @ComponentScan
5. @Component
6. @Value
7. Java Configuration
8. @Configuration
9. @Bean
10. @PropertySource
11. Environment
12. Lifecycle methods
13. Runners
14. @ConfigurationProperties
15. Properties file
16. YAML File
17. Profiles
18. Lombok API

===========================
spring core chapter 1
===============
Core:
This module/chapter gives all rules and guidelines to work with spring container.
1. DI and IOC
============

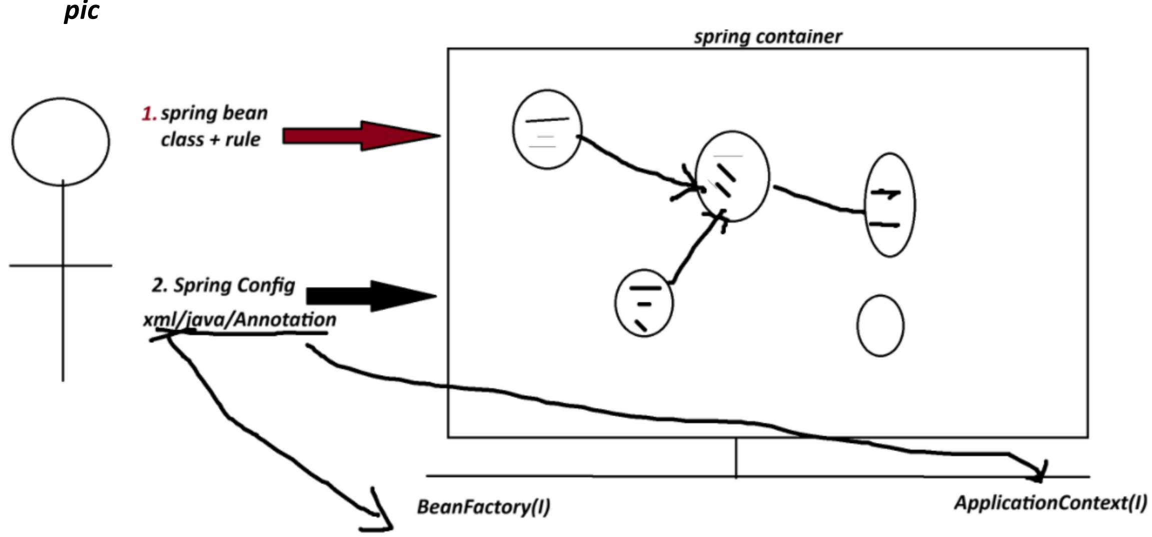
Spring Container :- It takes care of object.
1. Find/Scan our Classes(spring bean)
2. Create Object
3. Provide Data
4. Link Object
5. Destroy the Object

=>For this programmer has to provide two inputs-
1. Spring Bean : class+rules given by spring container.
2. Spring Configuration : This is main input cotains details like object name, relation with objects, data....etc.
xml/java/Annotation

*** Spring Container 2 Types:*
1. BeanFactory (Old container) : support only XML coonfiguration file.

2. ApplicationContext (New container) : support xml/java/Annotation configuration.

pic



=======================

Dependeny Injection (DI)

Theory	*programming*
OOPs	*java*
ORM	*HIbernate with JPA*
DI	*Spring IoC(Spring Container) -> Inversion of Control*

=> Depedency :- An instance variable (Field) exist inside a class (spring bean)

1. Primitive Type Dependeny (PTD) :- If a variable is created using (8+1) datatype.
(byte, short, int, long, float, double, boolean, char +string)
(Their wrapper classess too)

2. Collection Type Depedency (CTD) :- If a variable created using(4) types:
(java.util.package)
(List,Map,Set and Properties)

3. Reference Type Depedency (RTD) :- If a variable is creatd using other class/interface

Injection(4)
======
=> It give/provide/assign existed provide data to Depedency(Inside object)

1. Setter Injection (SI) :- provide data to variable using its set method(setter).

2. Constructor Injection(CI): provide data creating object using parameterized constructor.

3. LookUpMethod Injection(LMI) :- Spring bean Scope
[When parent bean is sigleton and child bean is prototype)

4. Interface Injection(II) : - Not Exist in Spring F/W.

```
class Employee{
   int id;
   String name;
   double sal;
}
```

id=~~0~~ 10
name=~~null~~ Raj
sal=~~0.0~~ 1200.0

Employee e=new Employee()

```
class Product{
   int id; //PTD
   String name; //PTD
   Boolean exist; //PTD

   List<String> models; //CTD
   Map<String,Integer> data; //CTD

   Vendor vob; //RTD
   Service sob; //RTD
}
```

```
class Vendor{}

interface Service{}
```

SESSION 5

===========

1. Spring Bean

2. STS Installation

Spring Bean

=========

-It is class that follow rules given by Spring Container.

Rule

-----

a. class must be public Type.

b. Recomended to keep in a package (basePackage Rule)

ex :

com.app.service

com.app.controller

c. Fields(Variables) are option, if required recomended to keep private.

private Integer empId;

private String empName;

d. If variable present in class,

=>define default constructor with setter/getter

=>(or/and) Parameterized Constructor.

ex:

```
public class Employee{
private Integer id;
private String name;
public Empoyee(){
}
 public void setId(Integer id){
this.id=id;
  }

public Integer getId(){
  return id;
  }
}
```

e. We can override Object(c) methods:

toString() equals() hashCode()

[because non-private, non-final,non-static]

f. Inheritance Rule:

Allowed ->Only Spring API is allowed (ie spring F/W classes and interface)

not allowed-> EJB, Servlet

g. Annotation Rule:

Only we can add Spring API Annotation and Inetegration API Annotations

(Hibernate with JPA, Restful webServices annotations);

ex:

@Component, @Service, @Repository.. etc.

STS Installation

=============

step 1:

go to official website

spring.io

link : - https://spring.io/tools#eclipse

step 2 download STS according your OS

step 3

Rigth click on ZIP file

click on "Extract Here"

step 4

Open extracted folder

Run : SpringToolsForEclipse

Final STEP STS installed successfully.

====================

session 6

=========

1. Spring Bean

Simple Java class that follow rules given by container.

2. Spring Configuration File(XML)

It gives object

details (name, data,links with other objects ... etc).

3. Test class=> Just find, container created or not? object is creatd or not?

===

Spring core XML Tags

===

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

---

1. To create Object

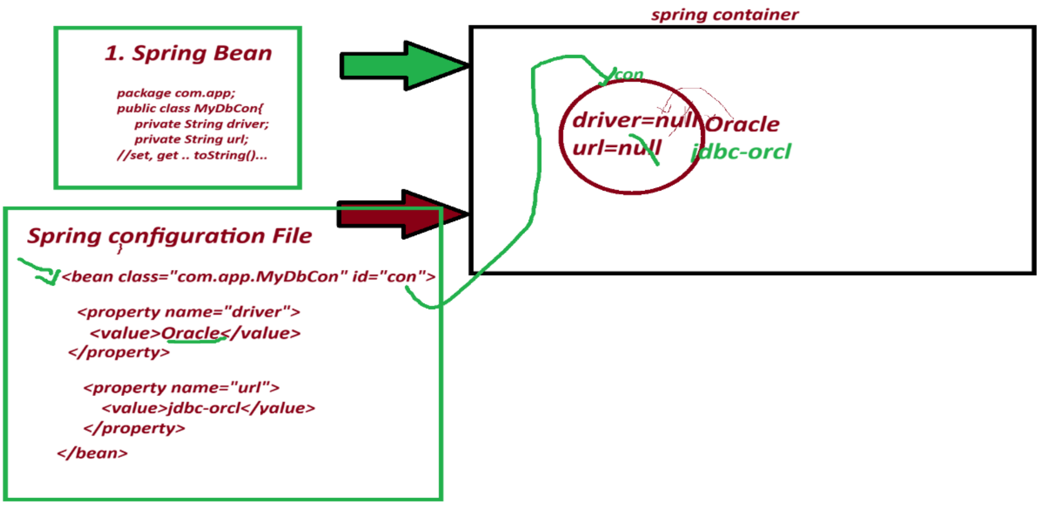
<bean id="objName" class="FullyQualifiedClassName"></bean>

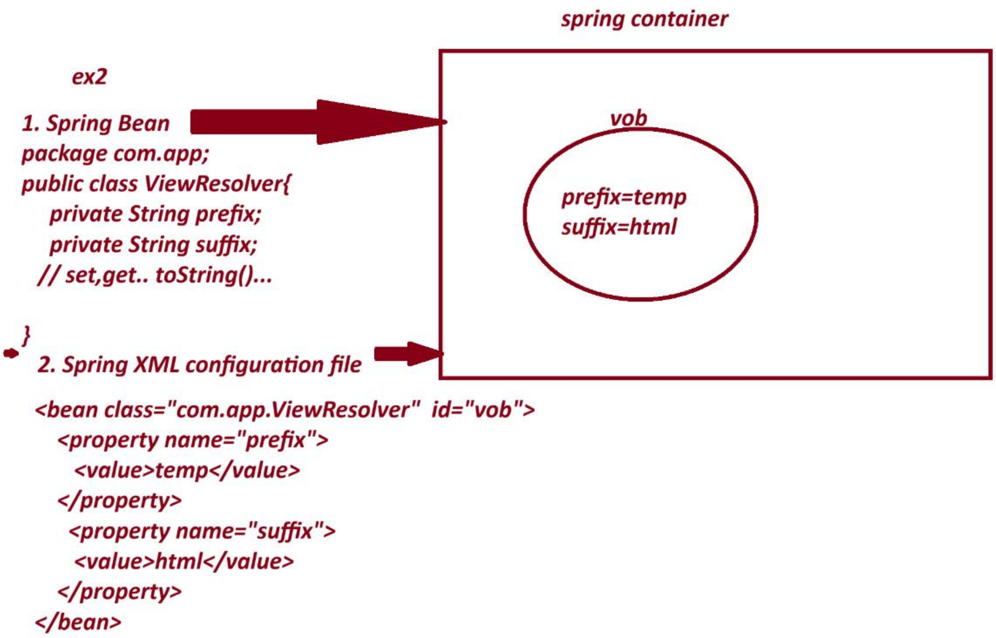
2. To call set method

<property name="variableName"></property>

3. To provide data to Primitive Variables

<value>data</value>





=================================

1. Spring bean

```
package com.app;

public class PaymentService{

    private int token;

    private boolean active;

    private String provider;

   //set.. set toString..etc

}
```

2. Spring XML configuration file.

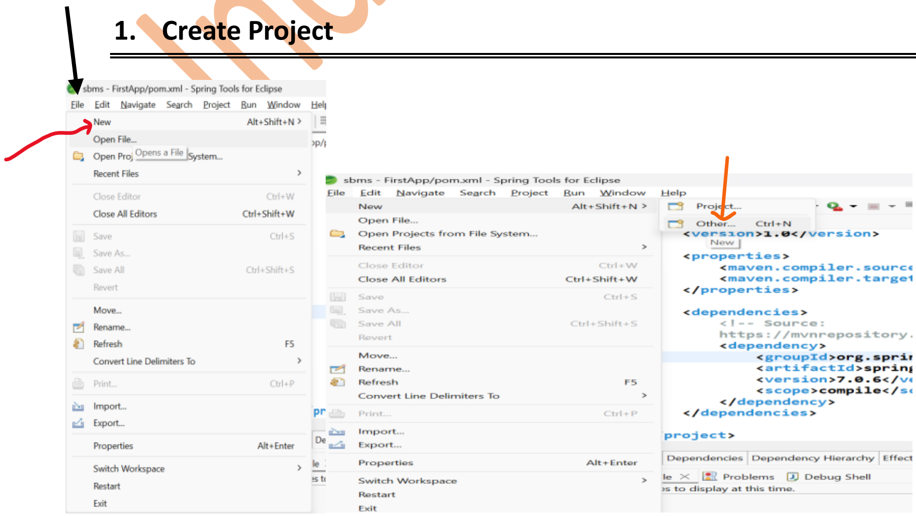
```
<bean class="com.app.PaymentService" id="psObj">
<property name="token">
<value>154678</value>
</property>
<property name="active">
<value>true</value>
</property>
<property name="provider">
<value>xyzBank</value>
</property>
</bean>
```

==

CREATE MAVEN PROJECT WITH BELOW STEP :-

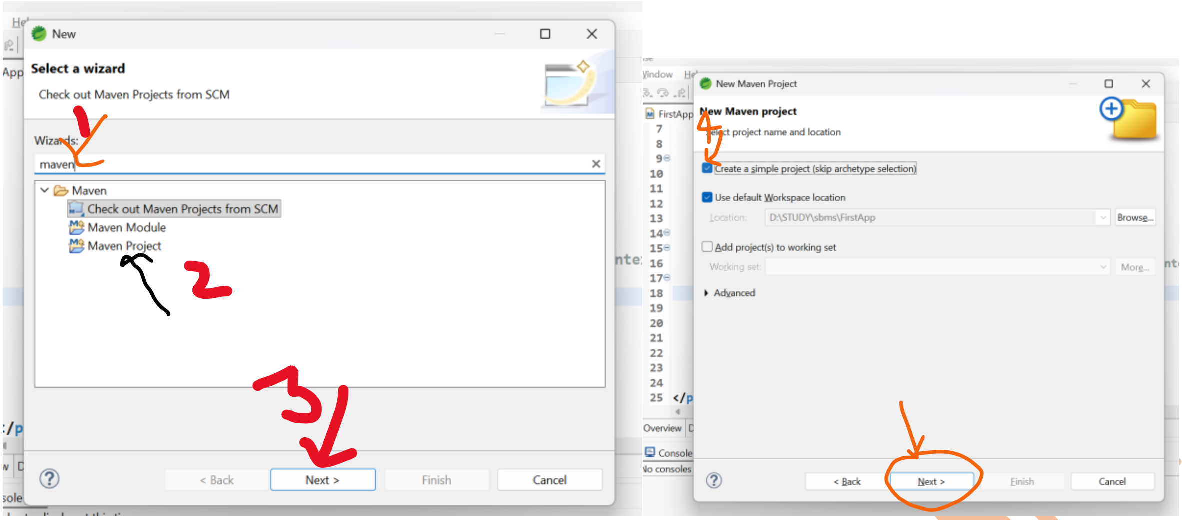
==

1. Create Project



File-> New-> Other-> Type maven->choose maven project-> next->

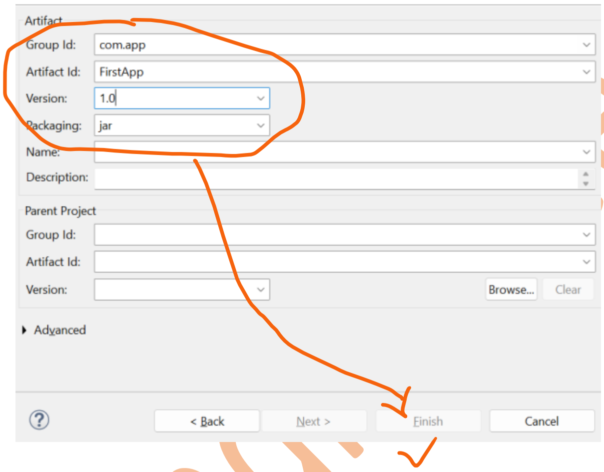
choose checkbox [v] create simple project -> next-> Enter details



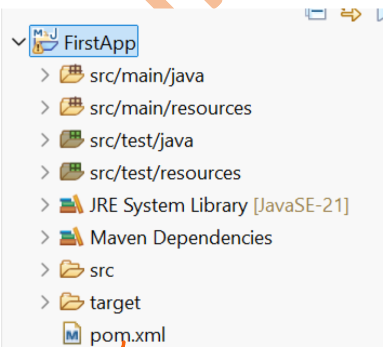
groupId : com.app

ArtficatId : FirstApp

version : 1.0



-> Finish



2. pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>

<artifactId>FirstApp</artifactId>

<version>1.0</version>

<properties>

<maven.compiler.source>21</maven.compiler.source>

<maven.compiler.target>21</maven.compiler.target>

</properties>


<dependencies>

        <!-- Source:

          https://mvnrepository.com/artifact/org.springframework/spring-context -->

  <dependency>

    <groupId>org.springframework</groupId>

      <artifactId>spring-context</artifactId>

      <version>7.0.6</version>

      <scope>compile</scope>

  </dependency>

</dependencies>

</project>
```

---

session 7

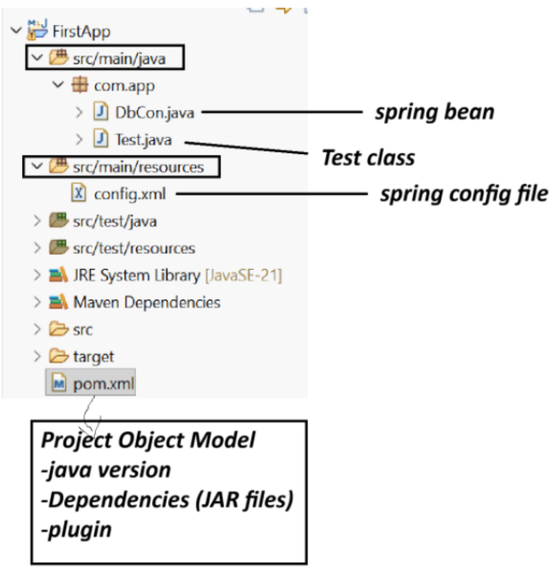
 ===========

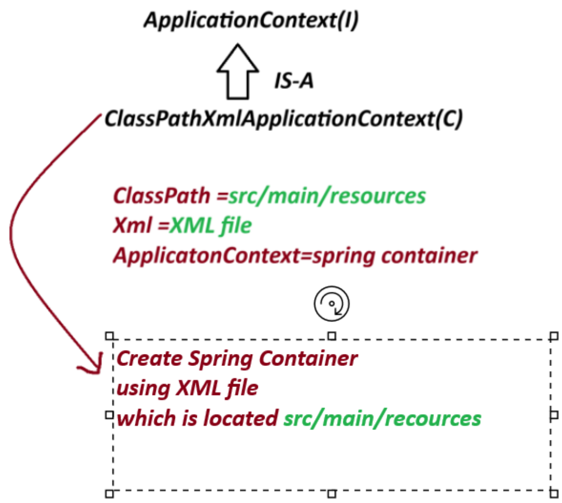
Spring Core First Application 3 Files.

1. Spring bean (.java)

2. Spring XML configuration (config.xml)

3. Test class(.java)





Note:-

1. Spring container are two Types.

(a) Legacy Container : BeanFactory(I) support only XML

(b) New Container : ApplicationContext(i) support all XML/java/Annotation config.

(c) ApplicationContext(I) : it is a interface.

we are using one of its Impl class:

ClassPathXmlApplicationContext(C).

===================================

Create Maven Project with below Step:

=================================

File>new>other>type maven>choose maven project>next>choose checkbox> create Simple project>next>Enter details

GroupId - com.app

ArtficatId - SecondApp

version - 1.0

----

code

SecondApp

=========

pom.xml

```
=============

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>SecondApp</artifactId>
<version>1.0</version>

<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

DbCon.java

```
package com.app;

public class DbCon {
  private String driver;
  private String url;
```

```
  //alt+shit+s
  public DbCon() {
super();
System.out.println("Constructor called");
  }
  public String getDriver() {
return driver;
  }
  public void setDriver(String driver) {
this.driver = driver;
System.out.println("Driver method called");
  }
  public String getUrl() {
return url;
  }
  public void setUrl(String url) {
this.url = url;
System.out.println("Url method called");
  }
  @Override
  public String toString() {
return "DbCon [driver=" + driver + ", url=" + url + "]";
  }

}
```

==

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean class="com.app.DbCon" id="mobj">
        <property name="driver">
        <value>Oracle Driver</value>
        </property>

        <property name="url">
        <value>JDBC-ORCL</value>
        </property>
    </bean>

</beans>
```

======================

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {
//ctrl+shit+O
public static void main(String[] args) {
ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
Object ob = ac.getBean("mobj");
System.out.println(ob);
}
}
```

----

output

Constructor called

Driver method called

Url method called

DbCon [driver=Oracle Driver, url=JDBC-ORCL]

---

session 8

=====================

Spring Container (DI - Depedency Injection)

=>Depedency - Variables exist in class

3 type

---

1. Primitive Type (8+1) : byte, short, int, long, float, double, boolean, char, String

2. Collection Type (4) : Set, List, Map and Properties

3. Reference Type (x) : using a class/interface variable

=> Injection : provide Data

- Setter Injection

- constructor Injection

-

LookUpMethod Injetion

- Field Injection (Annotations)

==

XML TAG

==

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

===

Collection Configuration

==

=> Spring container support working wiht collection configuration which are from java.util package

Collection Name	Collection Type	XML Tag Name
List	Interface	<list>
Set	Interface	<set>
Map	Interface	<map> and <entry>
Properties	Class	<props> and <prop>

List -> ArrayList

Set -> LinkedHasSet

Map -> LinkedHasMap

Properties --> NA

----

coding part collectionTypeS8

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>collectionTypeS8</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
```

```
</properties>

<dependencies><!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Spring bean

MyDetails.java

```
package com.app;

import java.util.List;
import java.util.Map;
import java.util.Properties;
import java.util.Set;

public class MyDetails {
//ctrl+shift+O
private List<String> data;
private Set<String> models;
private Map<Integer,String> design;
private Properties myinfo;

public MyDetails() {
super();
```

```
}

public List<String> getData() {
return data;
}

public void setData(List<String> data) {
this.data = data;
}

public Set<String> getModels() {
return models;
}

public void setModels(Set<String> models) {
this.models = models;
}

public Map<Integer, String> getDesign() {
return design;
}

public void setDesign(Map<Integer, String> design) {
this.design = design;
}
public Properties getMyinfo() {
return myinfo;
}public void setMyinfo(Properties myinfo) {
this.myinfo = myinfo;
}
@Override
public String toString() {
```

```
return "MyDetails [data=" + data + ", models=" + models + ", design=" + design + ", myinfo=" + myinfo +
"]";

}

}
```

----

xml config file

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


<bean class="com.app.MyDetails" id="myinfo">

<property name="data">

<list>

<value>A</value>

<value>B</value>

<value>C</value>

<value>A</value>

</list>

</property>

<property name="models">

<set>

<value>M1</value>

<value>M2</value>

<value>M3</value>

<value>M1</value>

</set>

</property>
```

```xml
<property name="design">
<map>
<entry>
<key>
<value>101</value>
</key>
<value>AB</value>
</entry>
<entry>
<key>
<value>102</value>
</key>
<value>Suraj</value>
</entry>
<entry>
<key>
<value>103</value>
</key>

<null></null>
</entry>

</map>
</property>

<property name="myinfo">
<props>
<prop key="A1">B1</prop>
<prop key="A2">B2</prop>
<prop key="A3">B3</prop>
<prop key="A5"></prop>
</props>
```

</property>

</bean>

</beans>

Test File

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

public static void main(String[] args) {

ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
//Object ob = ac.getBean("myinfo");
MyDetails ob = ac.getBean("myinfo",MyDetails.class);
System.out.println(ob);
System.out.println(ob.getData().getClass().getName());
System.out.println(ob.getModels().getClass().getName());
System.out.println(ob.getDesign().getClass().getName());
System.out.println(ob.getMyinfo().getClass().getName());
}

}
```

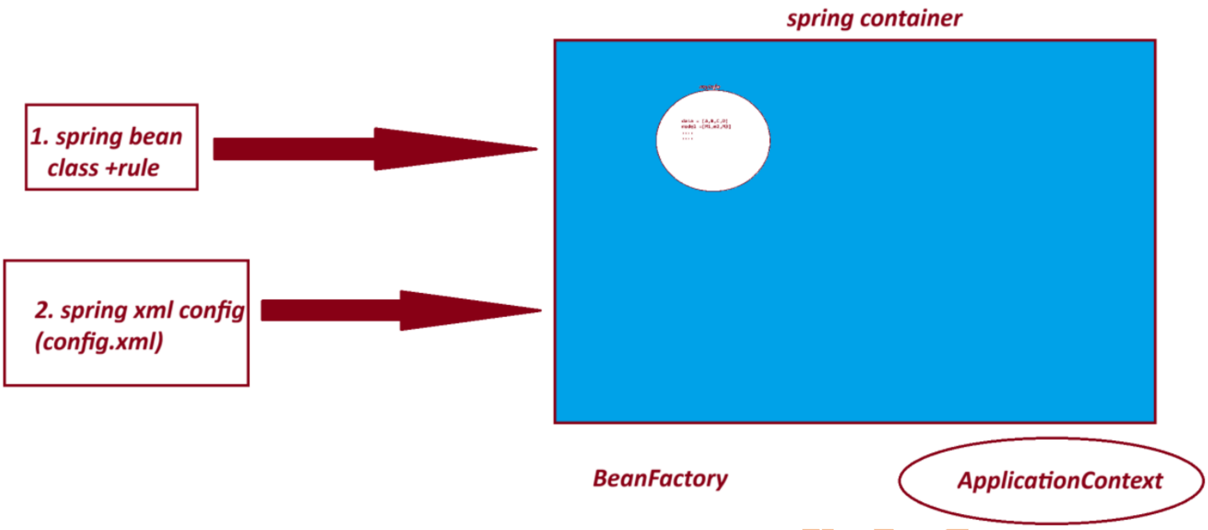
----

console output

```
MyDetails [data=[A, B, C, A], models=[M1, M2, M3], design={101=AB, 102=Suraj, 103=null},
myinfo={A1=B1, A2=B2, A3=B3, A5=}]
java.util.ArrayList
java.util.LinkedHashSet
```

java.util.LinkedHashMap

java.util.Properties



===

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========

3. Reference Type Depedency - Setter Injection.

============================

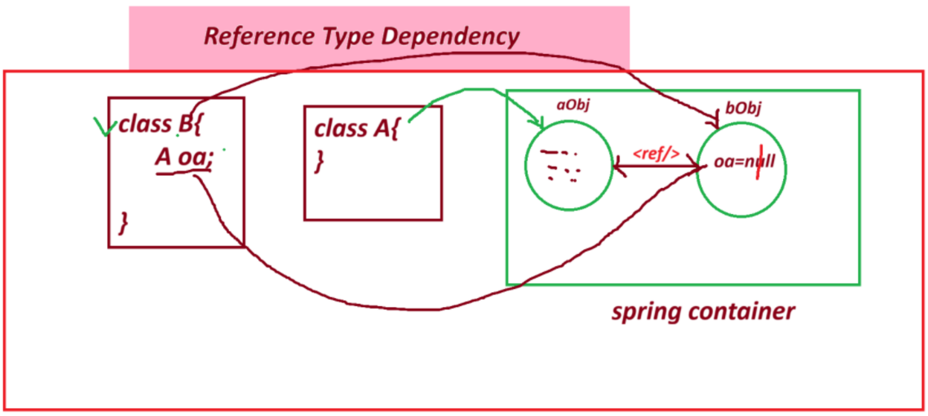
=> <ref/> : - tag is used to Link objects in case of HAS-A Relation.

Syntax:

<property name="variableName">

<ref bean="childObjName"/>

</property/>



======================

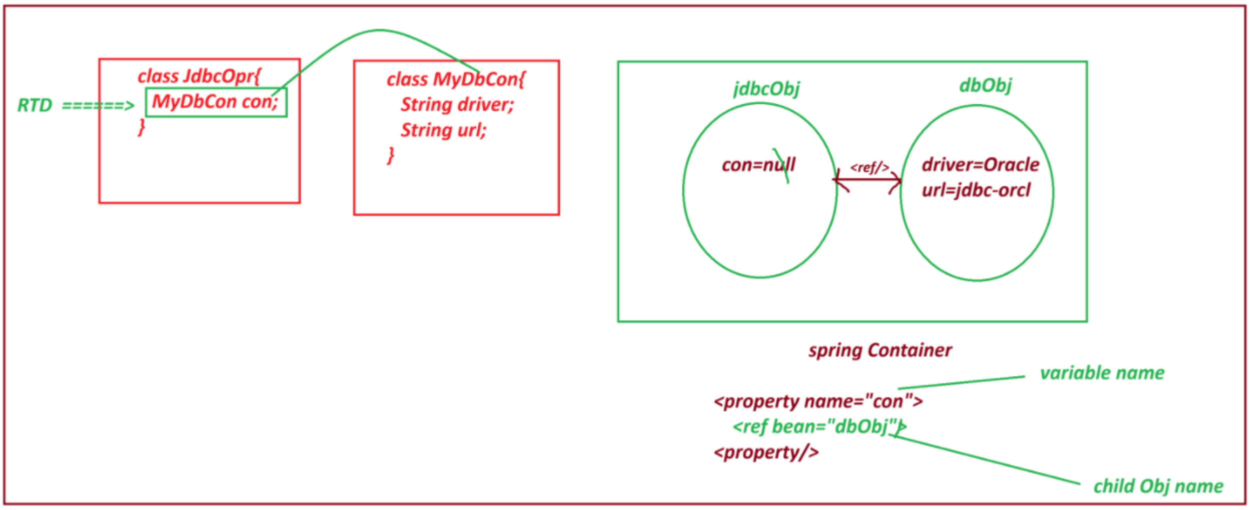
- If we do not use <ref/> then ref variable holds default value null.

=> spring container creates objet to both classes and link them by using <ref/> tag.

=> if child object nam (ref bean name) is not valid or not exist then spring container throws exception like

NoSuchBeanDefinitionException:

==



project : ReferenceTypeS9

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ReferenceTypeS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>
<dependencies>
```

```
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>

</project>
```

JdbcOpr.java

```java
package com.app;

public class JdbcOpr {
  private MyDbCon con; //RTD

  public JdbcOpr() {
super();
// TODO Auto-generated constructor stub
  }

  public MyDbCon getCon() {
return con;
  }

  public void setCon(MyDbCon con) {
this.con = con;
  }
```

```
  @Override
  public String toString() {
return "JdbcOpr [con=" + con + "]";
  }

}
```

===============

MyDbCon.java

```
package com.app;

public class MyDbCon {
  private String driver;
  private String url;

  public MyDbCon() {
super();
// TODO Auto-generated constructor stub
  }

  public String getDriver() {
return driver;
  }

  public void setDriver(String driver) {
this.driver = driver;
  }

  public String getUrl() {
return url;
  }
```

```
  public void setUrl(String url) {
this.url = url;
  }

  @Override
  public String toString() {
return "MyDbCon [driver=" + driver + ", url=" + url + "]";
  }

  //alt+shift+s

}
```

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean class="com.app.MyDbCon"  id="dbObj" >
      <property name="driver">
      <value>OracleDriver</value>
      </property>
       <property name="url">
       <value>JDBC-ORCL</value>
       </property>
  </bean>
   <bean class="com.app.JdbcOpr"  id="jdbcObj">
     <property name="con">
     <ref bean="dbObj"/>
```

```
    </property>
  </bean>


</beans>
```

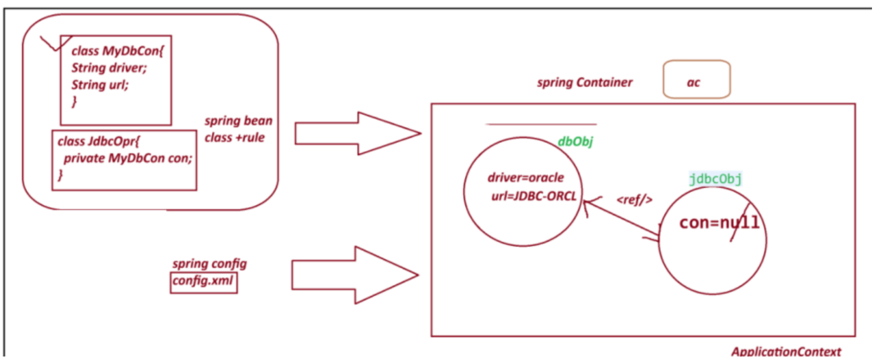
Test.java

```java
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {
public static void main(String[] args) {
ApplicationContextac=new ClassPathXmlApplicationContext("config.xml");
JdbcOpr jdbc = ac.getBean("jdbcObj",JdbcOpr.class);
System.out.println(jdbc);
}
}
```

output

JdbcOpr [con=MyDbCon [driver=OracleDriver, url=JDBC-ORCL]]

===



SI - Setter Injection :

Spring container creates object to given class using default constructor and provide data using setter method.

Employee emp=new Employee(); =======> <bean id="emp" class="Employee"/>

emp.setEmpId(10); =======> <property name="empId" value="10"/>

================================

*** Constructor Injection CI ***********

=================================

=> Sprint container creating object and providing data using Parameterized constructor.

Here , we use <constructor-arg> tag to indicate.

Employee emp=new Employee(10, "A"); =====>

```
<bean>
  <constructor-arg>
 <value>10</value>
</constructor-arg>
<constructor-arg>
 <value>A</value>
</constructor-arg>
</bean>
```

Note:

A. If a class has 3 param constructor then we need to pass <constructor-arg> 3 times in xml config file.

B. Data must be passed in order using <constructor-arg> as per constructor difined in class.

==

code for CI (constructor Injection)

----

pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ConstructorInjectionS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>

</project>
```

Product.java

```java
package com.app;

import java.util.List;

public class Product {

private String pcode;
```

```
private String model;

private List<String> data;

private Vendor vob;


public Product(String pcode, String model, List<String> data, Vendor vob) {

super();

this.pcode = pcode;

this.model = model;

this.data = data;

this.vob = vob;

}

@Override

public String toString() {

return "Product [pcode=" + pcode + ", model=" + model + ", data=" + data + ", vob=" + vob + "]";

}


}
```

Test.java

```
package com.app;


public class Vendor {


private String vcode;

private String vaddr;


public Vendor(String vcode, String vaddr) {

super();

this.vcode = vcode;

this.vaddr = vaddr;

}
```

```
@Override
public String toString() {
return "Vendor [vcode=" + vcode + ", vaddr=" + vaddr + "]";
}


}
```

----

config.xml

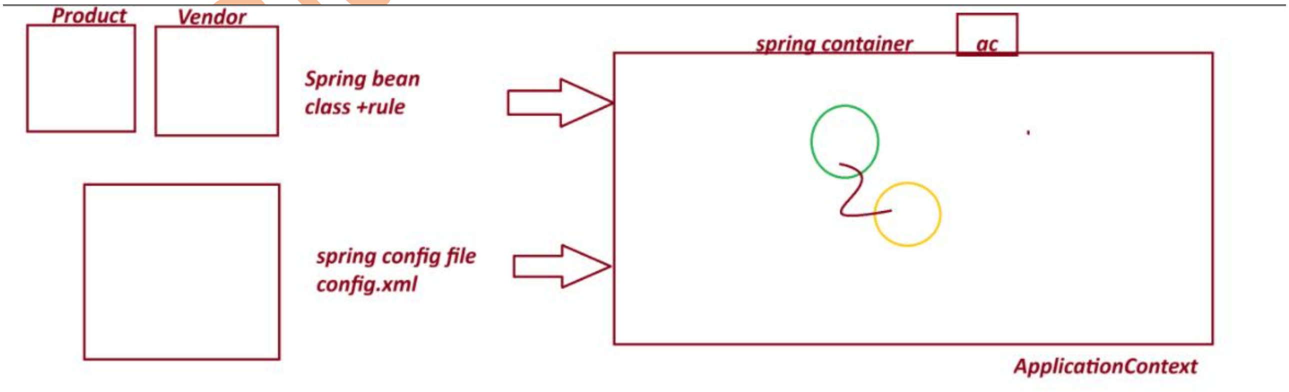
```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
       http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


 <bean class="com.app.Vendor" id="v1">
   <constructor-arg>
   <value>v1109</value>
   </constructor-arg>
   <constructor-arg>
   <value>IND</value>
   </constructor-arg>
 </bean>


 <bean class="com.app.Product" id="pobj">
  <constructor-arg>

<value>P01</value>
 </constructor-arg>



  <constructor-arg>
```

```
<value>HP-23444</value>
 </constructor-arg>


<constructor-arg>
<list>

<value>AA</value>

<value>BB</value>
</list>
</constructor-arg>


<constructor-arg>
<ref bean="v1"/>

</constructor-arg>
</bean>


</beans>
```

output

=========

Product [pcode=P01, model=HP-23444, data=[AA, BB], vob=Vendor [vcode=v1109, vaddr=IND]]



====================================

session 10

===========

Spring Annotation Configuration

This is Easy to Configure

Faster in execution compare to xml.

======================

1. Stereotype Annotation:

These annotation informs Spring Container to create Object.

They are 5 Annotation given by Spring Framework.

@Component : Create object inside container.

@Repostitory : Creates object + Database operations

@Service : Creates object + Transaction/logic/calculation..etc

@Controller : Creates object + Http Protocol (Web App)

@RestController : Creates object + Http Protocol (webservices)

@ComponentScan

@ComponentScans

2. Data Annotation/ Basic Annotation

@Value Annotation is used to provide data (field inject) to variable.

It has 3 usages:

1. Hard coding direct value to a variable.

2. Reading Data from Properties/YAML Files

3. - SpEL (Spring Expression language)

@Autowire

@Qualifier

@Primary

==============

ExcelExportAnnotationConfigS10

code

========

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ExcelExportAnnotationConfigS10</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Spring bean + annotation cofiguration

ExcelExport.java

```
package com.app;
import org.springframework.beans.factory.annotation.Value;
```

```
import org.springframework.stereotype.Component;
@Component("excel")
public class ExcelExport {
@Value("TEST")
   private String exportName;
@Value(".csv")
   private String fileExt;

@Override
public String toString() {
return "ExcelExport [exportName=" + exportName + ", fileExt=" + fileExt + "]";
}

}
```

MyAppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.ComponentScans;

@ComponentScan("com.app")
public class MyAppConfig {

}
```

Test.java

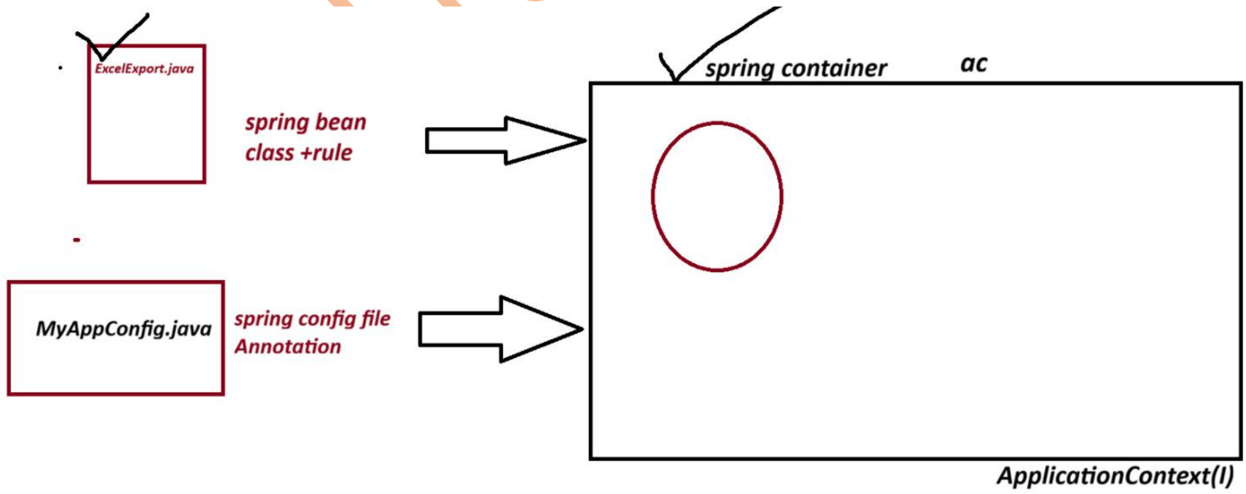
```
package com.app;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;
```

```
import org.springframework.context.ApplicationContext;


public class Test {
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(MyAppConfig.class);
	// Find classes from a package(basePackage)
//ac.scan("com.app");
	// object creation
	//ac.refresh();



	ExcelExport ee = ac.getBean("excel",ExcelExport.class);
	System.out.println(ee);
}
}
```

output

ExcelExport [exportName=TEST, fileExt=.csv]

Diagram



===========

session 11

========

Spring Annotation Configuration:

1. @Component : create Object to given class

2. @Value : To provide data (Field Injection) to variable

3. @ComponentScan : To find classes using basepakage.

====================Prorperties File=================

__________.properties

=> This is also caleld as key-val pair input as Text format(String)

=> This file is used to provide setup data to application.

=> Here, we can read data from properties file into variable using @Value("${key}")

=>Programmer has to define and load Properties file.

Use @PropertySource annotatio to load file, into spring container.

=> container creates Environment(I) Object (internally used StandardEnvironment(c)).

=> we can read data using @Value("${keyName}")

=> if we define duplicate key with different value (not recomened then last combination is taken into Container)

================

myapp.properties

#Symbol allowed for comment '#'

#key=value

#Symbol allwed to write keyName (dot(.), dash(-), underscore(_))

my.driver=oracle

my.url=oracle-jdbc

my.username=system

my.password=root

my.port=3306

my.active=fALSE

=============

my.port=4444 will be printed

=================

=> For boolean datatype valid input are : false and true

(case-insensitive, False,FASLE,FalSE, True,TRue etc)

========

Project Name: database-connectionS11

Example

code

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>database-connectionS11</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Test.java

```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;
```

```
public class Test {


public static void main(String[] args) {

ApplicationContext ac=new AnnotationConfigApplicationContext(MyAppConfig.class);

DatabaseCon obj = ac.getBean("databaseCon",DatabaseCon.class);

System.out.println(obj);

}


}
```

DatabaseCon.java

```
package com.app;


import org.springframework.beans.factory.annotation.Value;

import org.springframework.stereotype.Component;


@Component

public class DatabaseCon {


@Value("${my.db}")

private String driver;


@Value("${my.url}")

private String url;


@Value("${my.username}")

private String username;


@Value("${my.password}")

private String password;
```

```
@Value("${my.port}")

private int port;


@Value("${my.enable}")

private boolean active;


@Override

public String toString() {

return "DatabaseCon [driver=" + driver + ", url=" + url + ", username=" + username + ", password=" +
password

+ ", port=" + port + ", active=" + active + "]";

}



}
```

MyAppConfig.java

```
package com.app;


import org.springframework.context.annotation.ComponentScan;

import org.springframework.context.annotation.PropertySource;


@ComponentScan("com.app")

@PropertySource("classpath:myapp.properties")

public class MyAppConfig {


}
```

----

myapp.properties

```
# comment line

# symbol allowed to write keyName (dot . dash - underscore _)

#key=value

my.db=oracle
```

my.url=oracle-jdbc

my.username=indiafreeinternship

my.password=root

my.port=3306

#my.port=3345

#my.port=4444

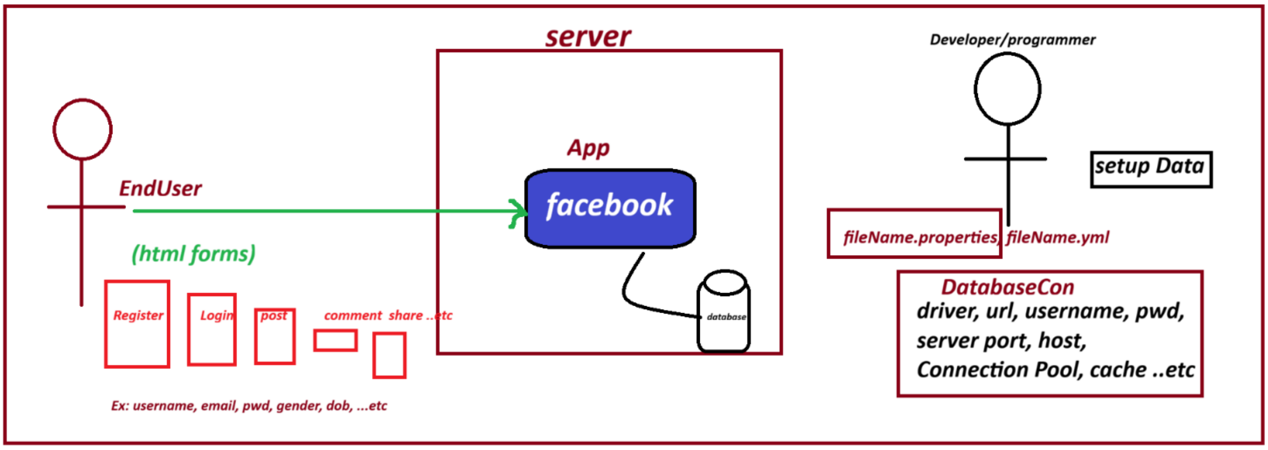
my.enable=tRUe

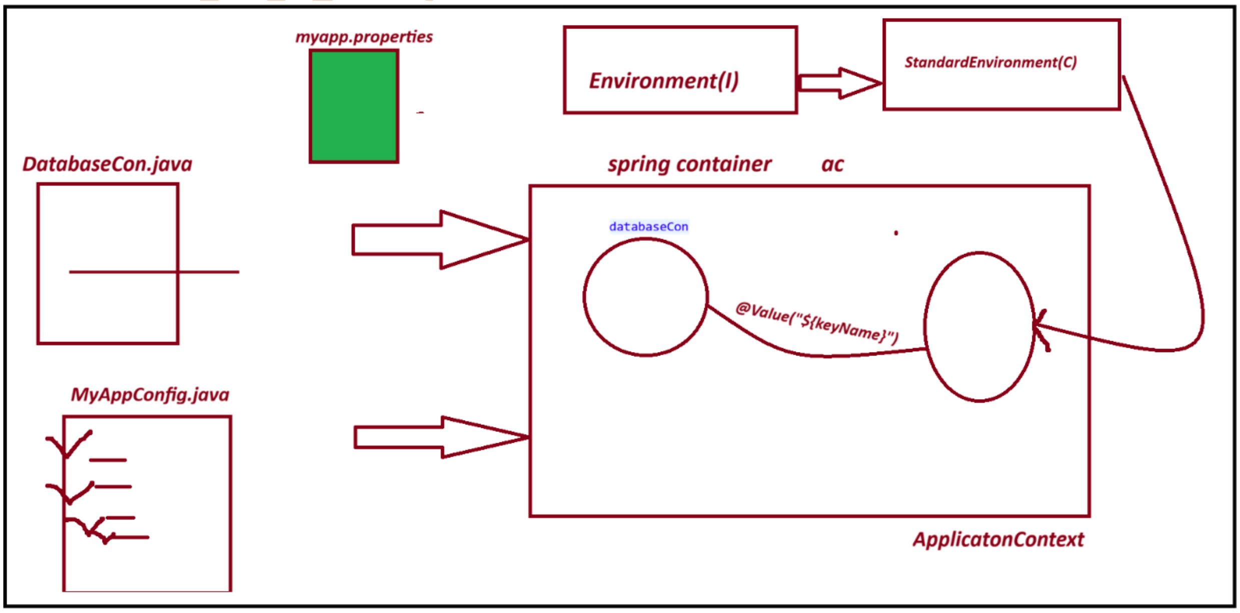
----

output

DatabaseCon [driver=oracle, url=oracle-jdbc, username=indiafreeinternship, password=root, port=4444, active=true]

Diagram





session 12

=======

Spring Annotation configuration

@Component Annotation : To create Object

@Value : Direct value to a variable / Read data from properties

@ComponentScan : back-Package to scan classes

@PropertySource : To Load Properties File

Autowiring

@Autowired Annotation

=> It is a process of Linking objects based on Association Mapping (HAS-A)

=> @Autowired Annotation is used. So, that container can link them.

=>Autowired Annotation Injects Dependecny Beans into Dependent bean.

=> In Case of XML Configuration <ref/> tag is used to link objects.

==

1. @Autowired will search for child class object.

2. If one child object is found then it is injected.

3. If Zero child object are found then throw exception NoSuchBeanDefinitionException, message like:

At least 1 bean required

==================

---------code----------

project name : AutowiredAnnoS12

=======================

pom.xml

<project xmlns="http://maven.apache.org/POM/4.0.0"

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>

<artifactId>AutowiredAnnoS12</artifactId>

```xml
<version>1.0</version>

<properties>

<maven.compiler.source>8</maven.compiler.source>

<maven.compiler.target>8</maven.compiler.target>

</properties>


<dependencies>

<!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-context</artifactId>

<version>7.0.5</version>

<scope>compile</scope>

</dependency>

</dependencies>



</project>
```

----

2. Spring bean

EmployeeDao.java

```java
package com.app;

import org.springframework.beans.factory.annotation.Value;

import org.springframework.stereotype.Component;

//@Component("edao")

@Component("edao")

public class EmployeeDao {
```

```
@Value("ORACLE")
private String dbName;

@Override
public String toString() {
return "EmployeeDao [dbName=" + dbName + "]";
}

}
```

----

3. spring bean

EmployeeService.java

```
package com.app;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;
//ctrl+shift+O
@Component("esobj")
public class EmployeeService {

//@Autowired(required=true)
@Autowired(required=false)
private EmployeeDao dao; // HAS-A

@Override
public String toString() {
return "EmployeeService [dao=" + dao + "]";
}
```

}

AppConfig.java

```
package com.app;

import org.springframework.context.annotation.ComponentScan;

@ComponentScan("com.app")
public class AppConfig {

}
```

Test.java

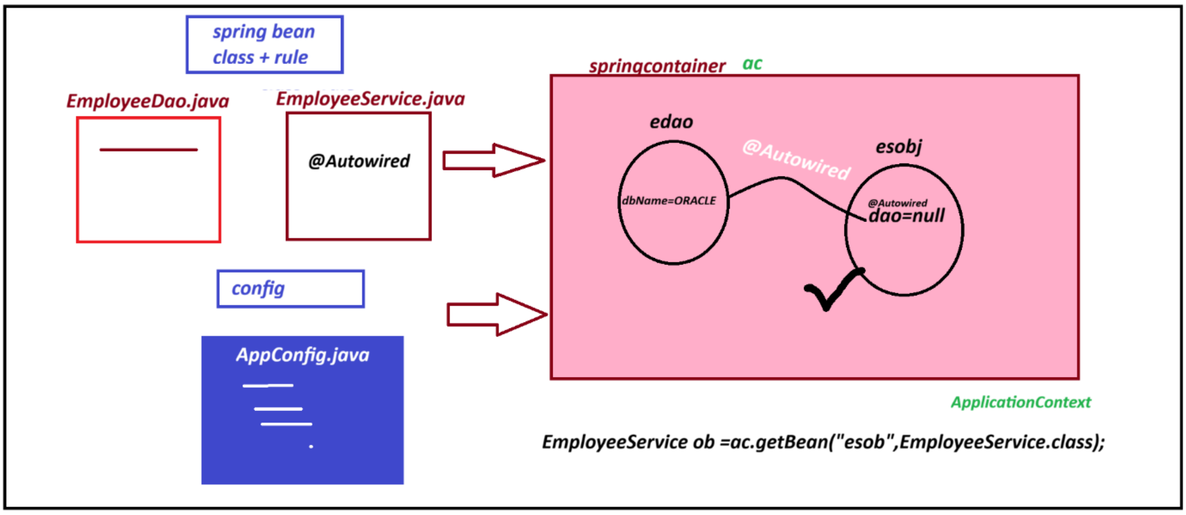
```
package com.app;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class Test {

public static void main(String[] args) {
ApplicationContext ac=new AnnotationConfigApplicationContext(AppConfig.class);
EmployeeService ob = ac.getBean("esobj",EmployeeService.class);
System.out.println(ob);

}

}
```

Output

EmployeeService [dao=EmployeeDao [dbName=ORACLE]]

===

Diagram



============

Sessino 13

Spring Bean -LifeCycle Methods

=====================

LifeCycle Methods : If a method is called by container/runtime in betwenn Object creation and destroy.

Spring F/W : 2 LifeCycle Methods (they are optional)

1. init-method : called after creating object

2. destroy-method : called before Distorying object

=======================================

=========init===========   ========destroy=====

File(data) -----> open file----------> Read data--------variable-> close File

===============================

We can implement LifeCycle Methods in 3 way

1. xml cofiguration

<bean ..... init-method="" destroy-method="">

=>we can define methods with any name, using below syntax:

public void <method_name>(){

//logic....

}

2. spring Lifecycle Interface:

=> Spring f/w has given 2 interfaces to implement lifecycle methods.

They are:

A. InitializingBean(I)

afterPropertiesSet()

B. DisposableBean(I)

Destroy()

=> Your Spring bean can implement above interface and override methods.

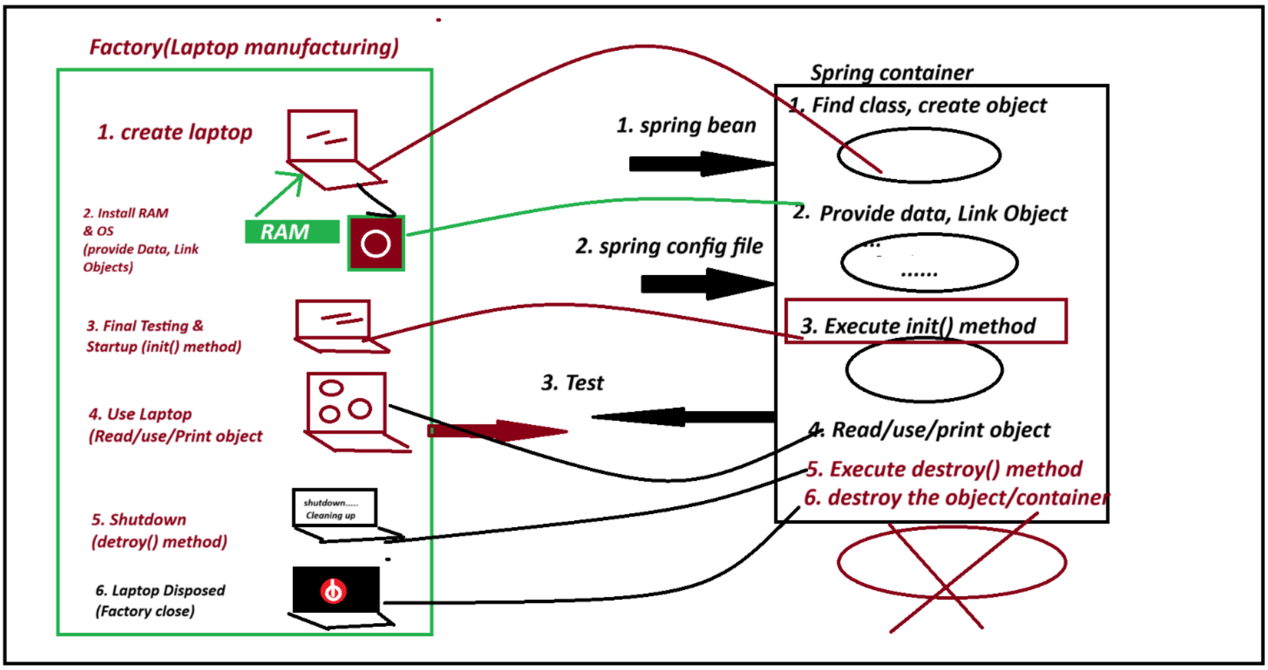
They are called by spring container.

3. JSR-Annotation : These are generic annotation Given by Sun/oracle .

a. @PostConstruct

b. @PreDestroy

diagram



===========code===============

1. xml cofiguration

```
<bean ..... init-method="" destroy-method="">
```

=>we can define methods with any name, using below syntax:

```
public void <method_name>(){
//logic....
}
```

==================code==================

project :- SpringBeanLifeCycleXMLS13

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.app</groupId>
  <artifactId>SpringBeanLifeCycleXMLS13</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <properties>
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:
https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
```

</dependencies>

</project>

----

spring bean

ExcelExport.java

```
package com.app;

public class ExcelExport {

private String fileName;
private String fileExt;

public ExcelExport() {
super();
System.out.println("from constructor");
}

public String getFileName() {
return fileName;
}

public void setFileName(String fileName) {
this.fileName = fileName;
System.out.println("from setter method");
}

public String getFileExt() {
return fileExt;
}

public void setFileExt(String fileExt) {
this.fileExt = fileExt;
```

```
}


public void setUp(){
System.out.println("3.FROM INIT METHOD");
}


public void clear() {
System.out.println("5.FROM DESTORY METHOD");
}


@Override
public String toString() {
return "ExcelExport [fileName=" + fileName + ", fileExt=" + fileExt + "]";
}


}
```

2. config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="
      http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">


<bean class="com.app.ExcelExport" id="exObj" init-method="setUp" destroy-method="clear">
<property name="fileName" value="ABC"/>
<property name="fileExt" value="CSV"/>
</bean>


</beans>
```

----

3. Test.java

```
package com.app;
```

```java
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

@SuppressWarnings("resource")
public static void main(String[] args) {
ClassPathXmlApplicationContext ac = new ClassPathXmlApplicationContext("config.xml");
ExcelExport ob = ac.getBean("exObj",ExcelExport.class);
System.out.println("FROM MAIN");
System.out.println(ob);
ac.registerShutdownHook();
//ac.close();
}

}
```

----

output

----

from constructor

from setter method

3.FROM INIT METHOD

FROM MAIN

ExcelExport [fileName=ABC, fileExt=CSV]

5.FROM DESTORY METHOD

============

2. spring Lifecycle Interface:

=> Spring f/w has given 2 interfaces to implement lifecycle methods.

They are:

A. InitializingBean(I)

afterPropertiesSet()

B. DisposableBean(I)

Destroy()

=> Your Spring bean can implement above interface and override methods.

They are called by spring container.

===========code==========

project Name :SpringBeanLifeCycleInterfaceS13

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.app</groupId>

  <artifactId>SpringBeanLifeCycleXMLS13</artifactId>

  <version>0.0.1-SNAPSHOT</version>

  <properties>

<maven.compiler.source>8</maven.compiler.source>

<maven.compiler.target>8</maven.compiler.target>

</properties>


<dependencies>

<!-- Source:

https://mvnrepository.com/artifact/org.springframework/spring-context -->

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-context</artifactId>

<version>7.0.5</version>

<scope>compile</scope>

</dependency>

</dependencies>

</project>
```

ExcelExport.java

```
package com.app;
```

```
import org.springframework.beans.factory.DisposableBean;
import org.springframework.beans.factory.InitializingBean;

public class ExcelExport implements InitializingBean, DisposableBean{

private String fileName;
private String fileExt;

public ExcelExport() {
super();
System.out.println("from constructor");
}

public String getFileName() {
return fileName;
}

public void setFileName(String fileName) {
this.fileName = fileName;
System.out.println("from setter method");
}

public String getFileExt() {
return fileExt;
}

public void setFileExt(String fileExt) {
this.fileExt = fileExt;
}
```

```
@Override
public String toString() {
return "ExcelExport [fileName=" + fileName + ", fileExt=" + fileExt + "]";
}
@Override
public void afterPropertiesSet() throws Exception {
System.out.println("FROM INIT METHOD AFTER PROPERTIESSET()");

}

@Override
public void destroy() throws Exception {
System.out.println("from destroy method destroy()");

}


}
```

===============

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

<bean class="com.app.ExcelExport" id="exObj">
<property name="fileName" value="ABC"/>
<property name="fileExt" value="CSV"/>
</bean>

</beans>
```

Test.java

```
package com.app;

import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

@SuppressWarnings("resource")
public static void main(String[] args) {
ClassPathXmlApplicationContext ac = new ClassPathXmlApplicationContext("config.xml");
ExcelExport ob = ac.getBean("exObj",ExcelExport.class);
System.out.println("FROM MAIN");
System.out.println(ob);
ac.registerShutdownHook();
//ac.close();
}

}
```

----

output

```
from constructor
from setter method
FROM INIT METHOD AFTER PROPERTIESSET()
FROM MAIN
ExcelExport [fileName=ABC, fileExt=CSV]
from destroy method destroy()
```

=======================

3. JSR-Annotation : These are generic annotation Given by Sun/oracle .

a. @PostConstruct
b. @PreDestroy

--------code--------------------

```
  <dependency>
<groupId>jakarta.annotation</groupId>
<artifactId>jakarta.annotation-api</artifactId>
<version>2.0.0</version>
</dependency>
```

----

project name : SpringBeanLifeCycleAnnotationS13

code

ExcelExport.java

```
package com.app;



import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

import jakarta.annotation.PostConstruct;
import jakarta.annotation.PreDestroy;

@Component("exObj")
public class ExcelExport {

@Value("SAMPLE")
private String fileName;
@Value(".csv")
private String fileExt;

public ExcelExport() {
super();
```

```java
System.out.println("from constructor");
}

@Override
public String toString() {return "ExcelExport [fileName=" + fileName + ", fileExt=" + fileExt + "]";
}



@PostConstruct
public void setUpA() {
System.out.println("FROM INIT METHOD");
}

@PreDestroy
public void clearB() {
System.out.println("FROM DESTROY METHOD");
}

}
```

Test.java

```java
package com.app;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class Test {

@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
ExcelExport ob = ac.getBean("exObj",ExcelExport.class);
System.out.println("FROM MAIN");
System.out.println(ob);
```

```
ac.registerShutdownHook();
//ac.close();
}

}
-----------
AppConfig.java
package com.app;

import org.springframework.context.annotation.ComponentScan;

@ComponentScan("com.app")
public class AppConfig {

}
------------
pom.xml
-------
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>SpringBeanLifeCycleXMLS13</artifactId>
<version>0.0.1-SNAPSHOT</version>
<properties>
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:
```

https://mvnrepository.com/artifact/org.springframework/spring-context -->

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-context</artifactId>

<version>7.0.5</version>

<scope>compile</scope>

</dependency>

<dependency>

<groupId>jakarta.annotation</groupId>

<artifactId>jakarta.annotation-api</artifactId>

<version>2.0.0</version>

</dependency>

</dependencies>

</project>

----

output

----

from constructor

FROM INIT METHOD

FROM MAIN

ExcelExport [fileName=SAMPLE, fileExt=.csv]

FROM DESTROY METHOD

===========

session 14

============

Spring - Java Configuration

==============

XML Configuration -- <bean> <property> <value> .. etc

Annotation configuration -- @Component, @Value, @Autowired .etc

java configuration -@Configuration @Bean

=======================

java configuration -@Configuration @Bean

1. Define one public class with any name (ex MyApp.java)

2. Apply @Configuration annother over class

3. For one object = define one method

```
@Bean
public PdfExport pObj(){
PdfExport p = new PdfExport();
p.setFileName("sbms");
p.setFileAuth("IndiaFreeInternship");
return p;

}
```

4. Add @Bean over method

----

1. Consider given class (spring bean) is a pre-defined class.

   can we use Annotation config?

   => No. We can't Annotation configuration.

   we can use annotation configuration we have source code only.

(_______.java) file

2. Can we xml/java configuration to define one bean for pre-defined class?

   yes Both support any type of class of configuration (pre-defined and user-defined)

3. what is the mean of @Configuration and @Bean?

   @Configuration says "i am java config file"

   @Bean - please create one object inside spring container

4. what is the diff b/w @Component and @Bean?

   @Component - used for programmer defined

   @Bean - Mainly for pre-defined class (support both pre-defined and user defined)

5. why we use Annotation configuration mostly

execution fast (short code)

in java configuration (lengthy code and slow execution)

================

example code

PdfExportJavaConfigS14

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.app</groupId>

  <artifactId>PdfExportJavaConfigS14</artifactId>

  <version>1.0</version>

  <properties>

<maven.compiler.source>8</maven.compiler.source>

<maven.compiler.target>8</maven.compiler.target>

</properties>


<dependencies>

<!-- Source:

https://mvnrepository.com/artifact/org.springframework/spring-context -->

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-context</artifactId>

<version>7.0.5</version>

<scope>compile</scope>

</dependency>

<dependency>

<groupId>jakarta.annotation</groupId>
```

```xml
<artifactId>jakarta.annotation-api</artifactId>
<version>2.0.0</version>
</dependency>

<!-- Source: https://mvnrepository.com/artifact/org.springframework/spring-jdbc -->
<dependency>
   <groupId>org.springframework</groupId>
   <artifactId>spring-jdbc</artifactId>
   <version>7.0.7</version>
   <scope>compile</scope>
</dependency>
</dependencies>
</project>
```

Test.java

```java
package com.app.test;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.app.bean.PdfExport;
import com.app.config.AppConfig;

public class Test {
@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
PdfExport ob = ac.getBean("pObj",PdfExport.class);

System.out.println(ob);

}
```

```java
}
-----------
PdfExport.java
package com.app.bean;

public class PdfExport {

private String fileName;
private String fileAuth;
public String getFileName() {
return fileName;
}
public void setFileName(String fileName) {
this.fileName = fileName;
}
public String getFileAuth() {
return fileAuth;
}
public void setFileAuth(String fileAuth) {
this.fileAuth = fileAuth;
}
@Override
public String toString() {
return "PdfExport [fileName=" + fileName + ", fileAuth=" + fileAuth + "]";
}



}
-----------
AppConfig.java
package com.app.config;
```

```
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;

import com.app.bean.PdfExport;

@Configuration
public class AppConfig {

// user defined
@Bean
public PdfExport pObj(){
PdfExport p = new PdfExport();
p.setFileName("sbms");
p.setFileAuth("IndiaFreeInternship");
return p;

}

}
```

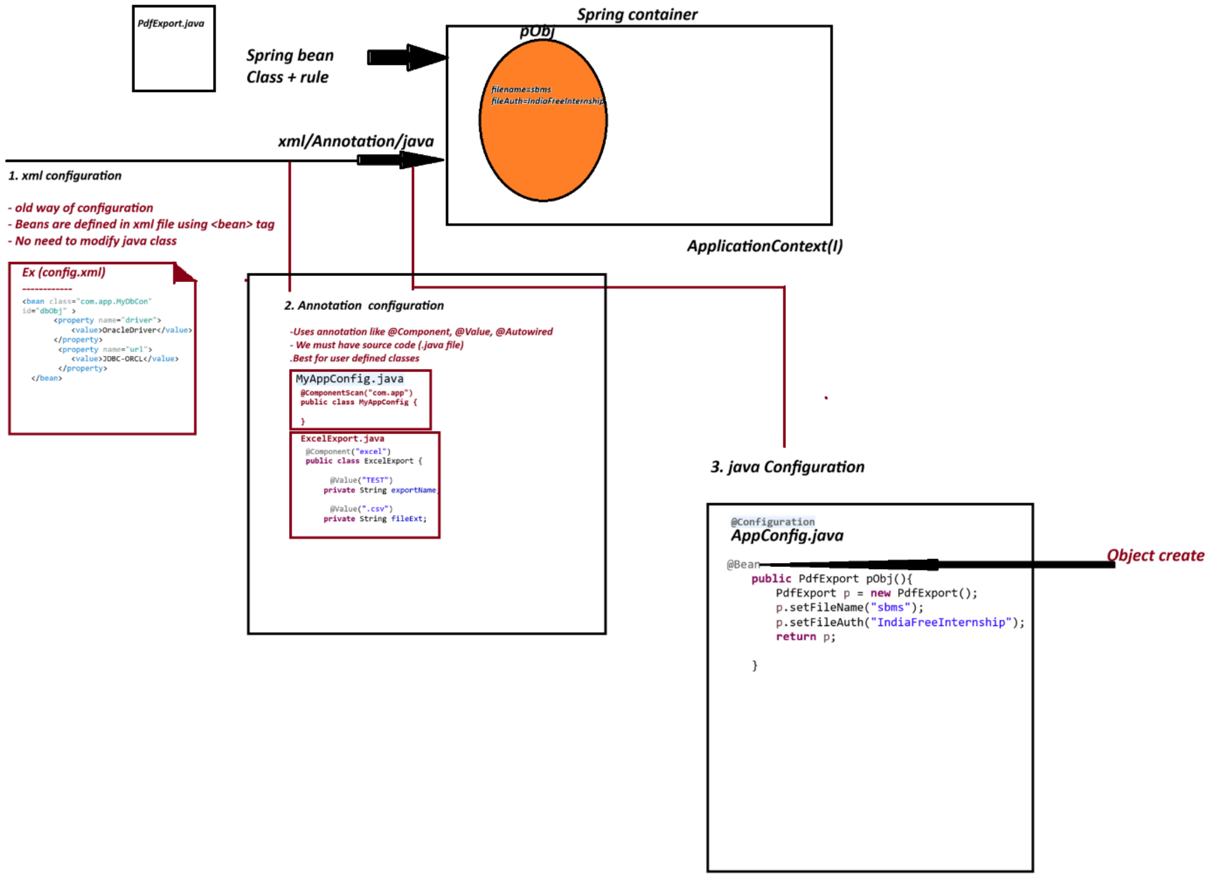
----

output

PdfExport [fileName=sbms, fileAuth=IndiaFreeInternship]

Diagram

============



Point	xml configuration	Annotation Configuration	Java Configuration
How to define Bean	<bean> tag in XML file	@Component, @Value, @Autowired	@Configuration class with @Bean tag
Need of Source Code	No	Yes(must have .java file)	No (Work for both)
Supports	Both pre-defined and user-defined classes	Only User-defined classes	Both pre-defined and user-defined classes
Multiple beans	Yes	Yes	Yes(easier)
Readability	Medium	High	High
Moder Approad	No	Yes(most used)	Yes

PRE-DEFINED CLASS
----
classes which are already available in libraries or framework.

we don't have source code. (.java)

ex:
JdbcTemplate
DriverManagerDataSource
RestTemplate
ArrayList(java API)

We Cannot apply @Component
(Cannot modify source code)

USER-DEFINED CLASS

Classes which are creatd by Programmer
We have source code (.java)

Ex
Student.java
Employee.java
Product.java
PdfExport.java

We can apply @Component
(We own the source code)

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=========

Java Configuration : Working with properties file

Lombok API

=============

Java Configuration : Working with properties file

=> properties file holds data in key=val

=> This is used to provide application setup data at runtime.

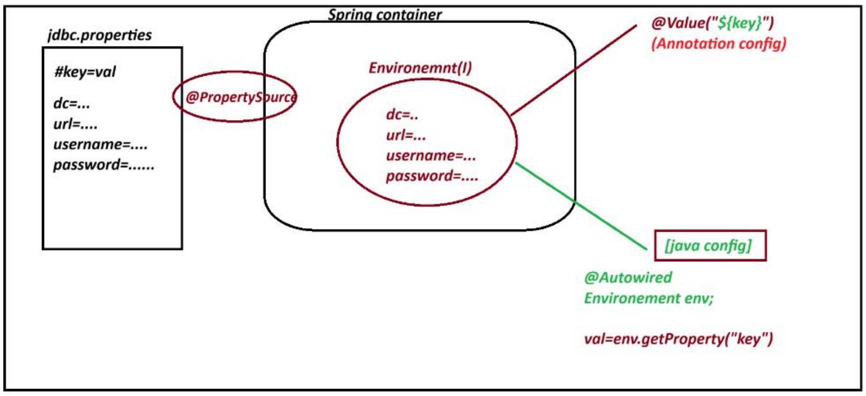
ex : Jdbc Connection details, ORM details, Email Config, Security View Resolvers ..etc

=> .properties file can be loaded using annotation @PropertySource

ex : @PropertySource("classpath:jdbc.properties")

=> Spring Container all these details into Environment(I) memory spring f/w has given impl class : StandardEnvironment(c)

=> This Environment(I) can be read from container to app using @Autowired.



Exmaple code

Dbcon-java-config-propertiesS15

----

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.app</groupId>

  <artifactId>Dbcon-java-config-propertiesS15</artifactId>

  <version>1.0</version>

  <properties>
```

```xml
<maven.compiler.source>8</maven.compiler.source>
<maven.compiler.target>8</maven.compiler.target>
</properties>
  <dependencies>
  <dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
<dependency>
<groupId>jakarta.annotation</groupId>
<artifactId>jakarta.annotation-api</artifactId>
<version>2.0.0</version>
</dependency>

  </dependencies>
</project>
```

```
---------------------------
MyDbConnection.java
-------
```

```java
package com.app.bean;

public class MyDbConnection {

private String driver;
private String url;
private String username;
private String password;


public MyDbConnection() {
```

```
super();
// TODO Auto-generated constructor stub
}
public String getDriver() {
return driver;
}
public void setDriver(String driver) {
this.driver = driver;
}
public String getUrl() {
return url;
}
public void setUrl(String url) {
this.url = url;
}
public String getUsername() {
return username;
}
public void setUsername(String username) {
this.username = username;
}
public String getPassword() {
return password;
}
public void setPassword(String password) {
this.password = password;
}
@Override
public String toString() {
return "MyDbConnection [driver=" + driver + ", url=" + url + ", username=" + username + ", password=" +
password+ "]";
}
```

}

AppConfig.java

```
package com.app.config;

import org.jspecify.annotations.Nullable;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.context.annotation.PropertySource;
import org.springframework.core.env.Environment;

import com.app.bean.MyDbConnection;

@Configuration
@PropertySource("classpath:jdbc.properties")
public class AppConfig {

@Autowired
Environment env;

@Bean
public MyDbConnection dbObj() {
MyDbConnection d = new MyDbConnection();
d.setDriver(env.getProperty("my.app.driver-class"));
d.setUrl(env.getProperty("my.app.url"));
d.setUsername(env.getProperty("my.app.username"));
d.setPassword(env.getProperty("my.app.password"));
return d;
}
```

Test.java

```
package com.app.test;


import org.springframework.context.annotation.AnnotationConfigApplicationContext;


import com.app.bean.MyDbConnection;
import com.app.config.AppConfig;


public class Test {


@SuppressWarnings("resource")
public static void main(String[] args) {
AnnotationConfigApplicationContext ac = new AnnotationConfigApplicationContext(AppConfig.class);
MyDbConnection ob = ac.getBean("dbObj",MyDbConnection.class);
System.out.println(ob);
}


}
```

----

jdbc.properties

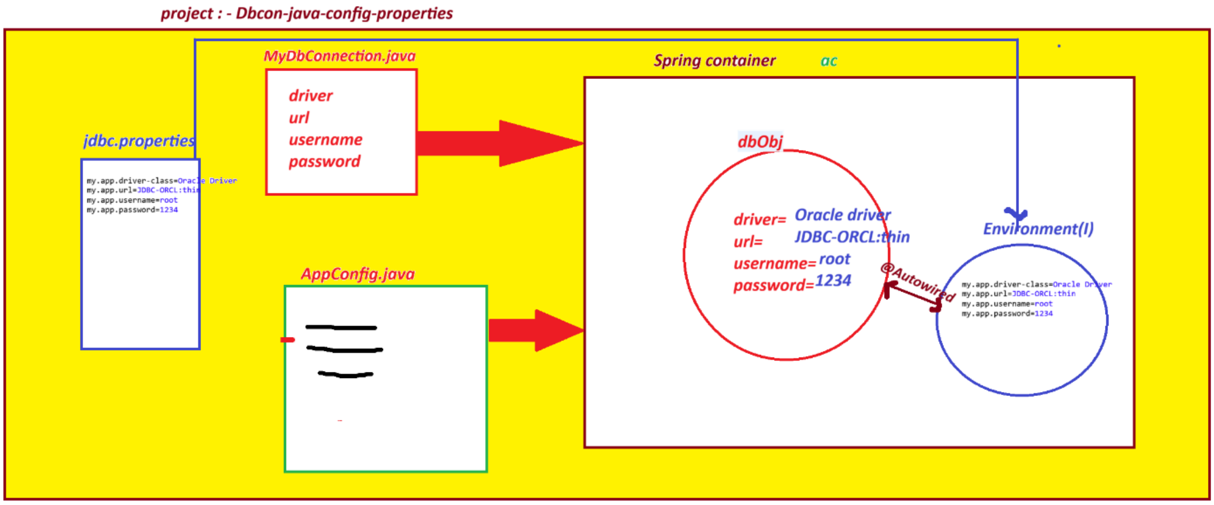
```
# key name can have A-Z a-z dash(-) dot(.) underscore(_)
my.app.driver-class=Oracle Driver
my.app.url=JDBC-ORCL:thin
my.app.username=root
my.app.password=1234


}
```

----

output

MyDbConnection [driver=Oracle Driver, url=JDBC-ORCL:thin, username=root, password=1234]

Diagram



====================

PROJECT Lombok API

=====================

=> It is open source Java API, used to generate code if we apply annotations

=> Lombok can generate code like

- Constructor (default, Parameterized)
- setter/getter methods
- toString()
- hashCode()
- equals()

===========

Annotation	Work
@Getter	Generate getter()
@Setter	Generate setter()
@ToString	toString()
@NoArgsConstructor	Default constructor
@AllArgsConstructor	Parameterzed constructor
@Data	Getter+Setter+toString+equals+hashCode

@Builder **Builder Patter implement**

@RequiredArgsConstructor **Required fields constructor**

----

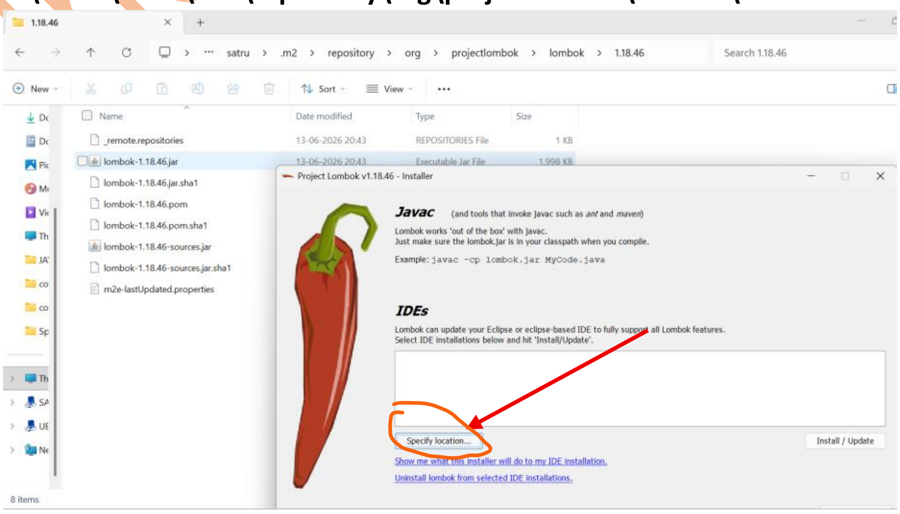
example:

```
import lombok.Getter;
import lombok.NoArgsConstructor;
import lombok.Setter;
import lombok.ToString;
@NoArgsConstructor
@Setter
@Getter
@ToString
public class MyDbConnection {

private String driver;
private String url;
private String username;
private String password;

}
```

=============

Go for Lombok Installation

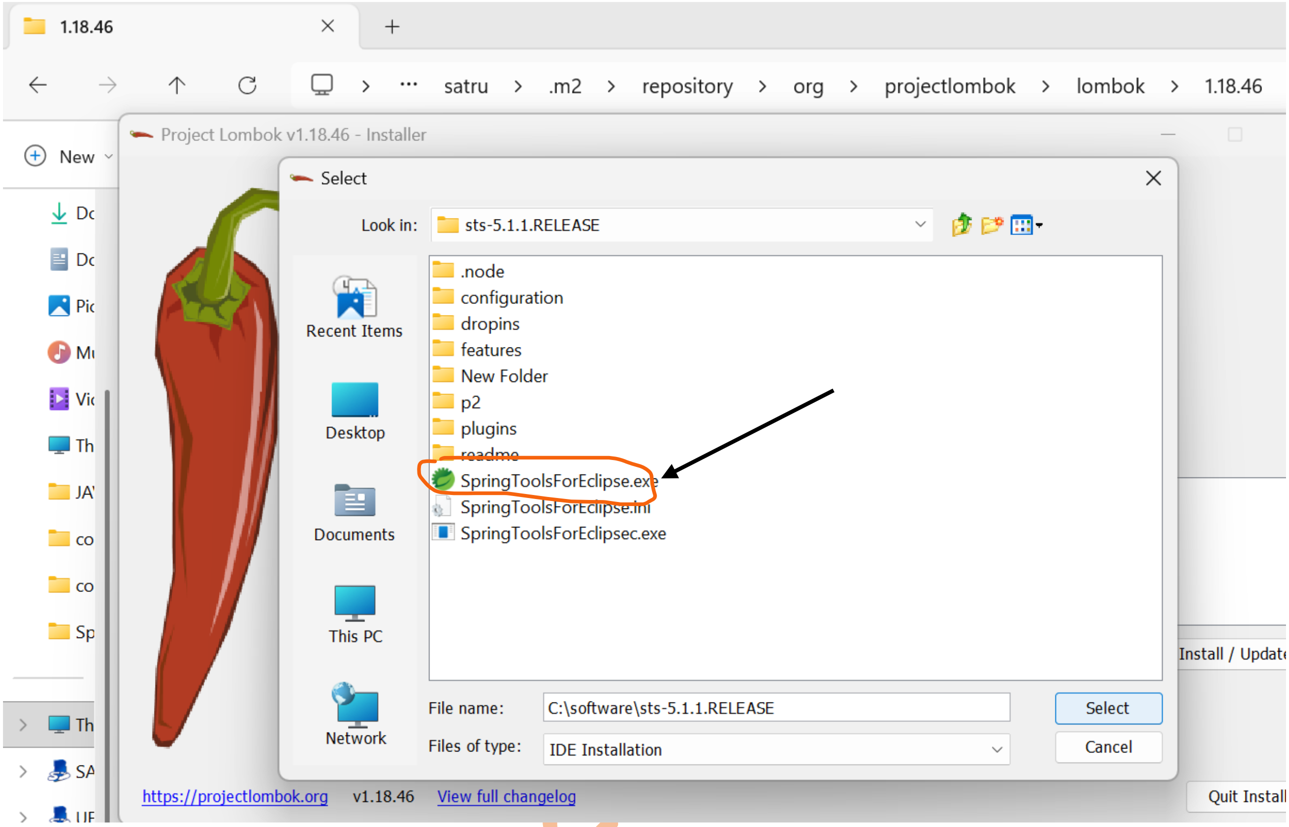
1. **C:\Users\satru\.m2\repository\org\projectlombok\lombok\1.18.46**



2. double click : lombok-1.18.46

3. wait sometime to detect STS IDE location

If not click on specifiy location and select sts IDE location



4. click on Install/Update > quite.

5. Open Again

=================

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===========